Human Performance and Perception of Uncertainty Visualizations in Geospatial Applications: A Scoping Review

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(Survey Paper)

***Abstract*—Geospatial data are often uncertain due to measurement, spatial, or temporal limitations. A knowledge gap exists about how geospatial uncertainty visualization techniques influence human factors measures. This comprehensive review synthesized the current literature on visual representations of uncertainty in geospatial data applications, identifying the breadth of techniques and the relationships between strategies and human performance and perception outcomes. Eligible articles described and evaluated at least one method for representing uncertainty in geographical data with participants, including land, ocean, weather, climate, and positioning data. Forty articles were included. Uncertainty was visualized using multivariate and univariate maps through colours, shapes, boundary regions, textures, symbols, grid noise, and text. There were varying effects, and no definitive superior method was identified. The predominant user focus was on novices. Trends were observed in supporting users understand uncertainty, user preferences, confidence, decision-making performance, and response times for different techniques and application contexts. The findings highlight the impacts of different categorizations within colour and shape techniques, heterogeneity in perception and performance evaluation, performance and perception mismatch, and differences and similarities between novices and experts. Contextual factors and user characteristics, including understanding the decision-maker's tasks, user type, and desired outcomes for decision-support appear to be important factors influencing the design of effective uncertainty visualizations. Future research on geospatial applications of uncertainty visualizations can expand on the observed trends with consistent and standardized measurement and reporting, further explore human performance and perception impacts with 3-dimensional and interactive uncertainty visualizations, and perform real-world evaluations within various contexts.**

***Index Terms*—Uncertainty, visualization techniques and methodologies, human performance, perception and psychophysics, human factors.**

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I. Introduction

RISK is inherent in decision-making tasks, where effective decision-makers assess risk using available information [1], often derived from environmental cues that change over time. However, available data are frequently uncertain—whether as estimations at a given moment or predictions of a future state. Importantly, uncertainty influences decision-making [1], and individuals process uncertain information subjectively, shaped by personal risk tolerances and experience.

In systems design, it is essential to account for potential biases when presenting uncertain data to users. Understanding these biases enables designers and developers to optimize how information is conveyed across different environments, ultimately affecting user interactions and decision-making outcomes. Increasing task complexity has led humans to rely more on computer algorithms [2], often without recognizing how this reliance may shape their interpretation and use of information [3]. A decision maker's expertise also influences how uncertainty is handled. Experienced individuals may integrate new information with prior knowledge to create mental models of uncertainty and make optimized decisions [4], [5]. They may also rely on intuition and heuristic shortcuts in familiar situations [6], [7], [8].

In contrast, novice decision-makers tend to depend on stereotypes [4], conscious effort, and rule-based reasoning, often struggling to manage unexpected uncertainty [6], [7], [8]. Given the potential for significant consequences in high-risk fields such as healthcare (e.g., medical diagnoses), finance (e.g., central bank interventions), and defence (e.g., adversary tracking), it is crucial to consider the human factors influencing decision-making. This is particularly relevant in designing computerized decision-support systems, where uncertainty visualization may play a key role.

As numerical indications of accuracy may be insufficient for effectively conveying uncertainty [9], visualizations offer additional context to help users assess potential risks and benefits, especially in situations of high uncertainty [10]. However, the effectiveness of uncertainty visualization depends on how the information is presented [4], [11], [12], [13]. Depending on the visualization technique used, decision-makers may be better equipped to interpret data, avoid misinterpretation, refine mental models, mitigate decision-making risks, and identify potential

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measurement errors, data quality issues, or inaccuracies [10], [12], [14], [15].

This review specifically examines the visualization of geospatial data and its unique uncertainty characteristics. Geospatial data are particularly prone to uncertainty due to measurement limitations (i.e., aliasing and biases), spatial constraints (i.e., coverage area, observation density, resolution) and temporal factors (i.e., when the data was collected) [16], [17], [18], [19]. For example, as climate change reshapes the Arctic into a potential underwater battlespace, Canada's defence research increasingly focuses on assessing the performance of acoustic sensors in these uncertain underwater environments. Situational awareness is critical for anti-submarine warfare missions, particularly in understanding a vessel's sensor detection range within its environment. Geospatial uncertainty in these contexts arises from several factors, necessitating thoughtful consideration of visualization approaches that support human factors.

First, uncertainty arises from the environment due to inadequate sampling that creates poor resolution both spatially and temporally [20], [21]. Specific factors include ambient noise, bottom topography, ocean temperature, salinity, and bottom type [22]. Second, uncertainty in the sonar data collected within this uncertain environment may be further compounded by limitations in the precision and accuracy of sensor instrumentation [21]. Finally, detection of a specific submarine threat—or counter detection of one's vessel—within this uncertain environment, with imperfect sensors, is influenced by the vessel's motion, acoustic signature, and tactics, each introducing additional layers of uncertainty [23], [24]. While models continue to be developed to evaluate and quantify geospatial uncertainty mathematically [25], [26], [27], it is becoming increasingly important to understand how to visualize uncertainty to account for the human factors involved in decision-making tasks. This includes understanding how to convey uncertainty that supports conceptualization and improves risk-based decision-making performance.

For the military and maritime context, few available literature reviews explore the impacts of various methods of representing data uncertainty and risk through geo-visualization and undersea representations [28], [29], despite the general increase in research conducted within this field since 1990 [13], [29]. The most recent review that includes information visualization with uncertainty discusses strategic management decisions from general cognitive psychology, education, finance, consumer research and human-computer interaction [30]. Other reviews comprehensively summarize the literature on visualizing uncertainty for specific and general geographical contexts [28], [29], including uncertainty representations with geospatial and healthcare data [29]. One review includes geospatial contexts and decision-making impacts [13], where the authors found evidence to support positive and negative outcomes. The results of prior reviews highlight that presenting uncertainty can affect understanding and decision-making, and there is a need to deeply explore the context in which uncertainty information is displayed. It remains to be understood which visualization techniques, settings, user expertise, and task scenarios may result in more (or less) desirable outcomes.

A comprehensive review is needed to examine the relationships between different approaches to uncertainty visualization and human factors-related outcomes to guide the design of interfaces in geospatial contexts and inform decisions on conveying uncertainty. The objectives of this review are to (1) synthesize existing knowledge on how various visualization techniques impact human factors measures in decision-making contexts and (2) identify future research directions for developing and evaluating new methods for representing information uncertainty to enhance human performance. The findings will provide valuable insights for display designers integrating uncertainty visualization into interfaces and highlight research gaps and challenges that require further exploration in geospatial applications.

II. Methods

A. Research Questions

The research questions for this review were collectively informed by the need to visualize the uncertainty of geospatial information in ways that support human understanding and decision-making, the limitations of current literature review outcomes, and preliminary searches for literature on uncertainty visualizations that included human subjects evaluations. The following research questions were iteratively developed through this process before the formal search of the literature was performed:

1) What methods are used to visualize uncertainty in geospatial data to human operators?
2) What are the impacts of these methods on human performance and perception (HPP) of uncertainty?
3) What knowledge gaps exist in understanding how to visualize geospatial uncertainty to improve HPP of uncertainty?

B. Article Relevancy

This review included the following databases, which were initially searched throughout November 2021 and June 2022, with the search being updated in January 2024: Association for Computing Machinery (ACM) Digital Library, Institute of Electrical and Electronics Engineers (IEEE), PsycINFO, Scopus, and Web of Science. Relevant articles were also identified through hand-searching references using the snowballing technique, a popular method in systematic literature reviews for software engineering [31]. The resulting articles from each search were exported to Covidence (Veritas Health Innovation Ltd.), where duplicates were automatically identified and removed.

The search strategy was developed for each database, using keywords related to four primary search categories: (1) visualization or representation, (2) uncertainty, (3) geospatial application context, and (4) human performance or perception factors. Detailed search strategies are provided in the supplementary material (Table S1). For Web of Science, a sample search strategy which resulted in 9974 articles is provided as an example:

ALL=((visual OR represent* OR communicat*) AND (uncertainty) AND (geographic OR geodata OR geospatial OR ocean OR weather OR climate OR undersea OR underwater OR sonar OR radar) AND (decision OR judg* OR reason* OR "problem solv*" OR "decision-making" OR "decision making" OR decide* OR perform* OR "human performance" OR "human perception" OR capab* OR behaviour* OR behavior* OR assess*))*

C. Article Eligibility

Eligibility criteria for article inclusion were guided based on information provided in previous reviews and the preliminary literature search performed before the formal search. Articles were eligible for inclusion in the full review if they met the following criteria: (1) describe at least one method or technique for visually representing geospatial data uncertainty and report on the results of quantitative or qualitative evaluations with human participants in at least one factor related to human factors; (2) are peer-reviewed articles or complete proceedings papers written in English. No limits were placed on the participants' age or the study protocol. Articles were excluded if they looked at applications beyond geospatial data, such as health and disease-related data, economics, and finance. Otherwise, articles were excluded if they were commentaries, editorials, books and book chapters, lectures and addresses, systematic reviews, scoping reviews, narrative reviews, dissertations, or conference abstracts.

D. Data Extraction & Charting

A standardized Excel (Microsoft Corporation) spreadsheet was iteratively developed to extract the following data from the included articles: author(s) and date, geographical location, research context and setting, research objective, study design, number and type of participants, type of uncertainty visualization method(s) or technique(s) used, uncertainty data being represented, interface dimension (i.e., two-, three-dimensional), human performance task or perception, objective and subjective metrics, performance and perception outcomes, key findings, and research gaps. The form was iterated during the full-text review process until it was determined that all relevant data was being extracted to provide a comprehensive review.

E. Data Collation & Summarization

Summary tables were employed to synthesize the extracted data. The results were first summarized using descriptive statistics regarding visualization techniques and environments. Tables were also constructed to visually display the reported impacts of the uncertainty representation techniques on the studied human factors metrics with an effect direction arrow [32], indicating the general trend of the technique on the factor, as interpreted from each study's results. Sometimes, comparisons were made between visualizations with and without uncertainty, between multiple uncertainty visualizations, or both. The individual trends were synthesized for each technique and within other categories by calculating averages and standard deviations after assigning a numerical value to individual effect direction arrows: -1 (↓), 0 (↔), 1 (↑). Individual studies with multiple evaluations or experiments for a single technique were averaged, resulting in effect direction values less than 1. While presenting the results of the various techniques on HPP factors qualitatively, this review provides a comprehensive summary that supports the quantitative findings of the various trends, highlighting research gaps on the potential performance and perception effects of uncertainty representations with fewer evaluations or conflicting evidence, which can be explored through further research.

F. Consultation With Defence Scientists

The preliminary results of this review were presented at the 2022 Human Systems Performance Maritime Annual Meeting, which was held virtually through the Technical Cooperation Program. This meeting brought together defence scientists from the five-nation forum—comprising the United Kingdom, United States, Canada, Australia, and New Zealand—to share ideas, coordinate research efforts, and enhance interoperability while avoiding duplication of work. Given this group's expertise and potential to apply or expand upon these findings, we considered their feedback and questions particularly valuable.

The session format consisted of oral presentations followed by five-minute question-and-feedback periods. Our presentation covered an overview of the review methodology, characteristics of the included studies, and preliminary findings on the various uncertainty visualization techniques explored in geospatial applications. At this stage of the review, the articles for inclusion had been identified, but the results had not yet been fully synthesized or analyzed. One key observation presented by the first author was a potential mismatch between perception and performance—specifically, cases where participants' preferred uncertainty visualization methods did not necessarily lead to improved decision-making.

A central question that emerged from the discussion was whether differences existed between expert and novice participants. As this factor had not been originally extracted, we decided to incorporate it into our study characteristics and analysis. We believe this addition enhances the comprehensiveness of our review and identifies additional areas for future research.

III. Results

The literature search identified 21,205 articles; two were identified with hand-searching and snowball sampling (Fig. 1). There were 5,591 duplicate records automatically removed by Covidence, and three were manually marked as duplicates. The first author reviewed the title and abstracts of the 15,611 articles remaining to determine whether the abstracts of eligible article types discussed details related to uncertainty visualization of geospatial data and human subjects evaluation. When the first author was uncertain whether the abstract met the inclusion criteria yet discussed uncertainty visualization techniques within a geospatial domain, the article was included in the full-text review. The number of articles after the abstract screening stage was 130, and the full text was read to confirm whether the eligibility criteria were met. The full-text review initially resulted in 35 articles that met the eligibility criteria. The excluded articles

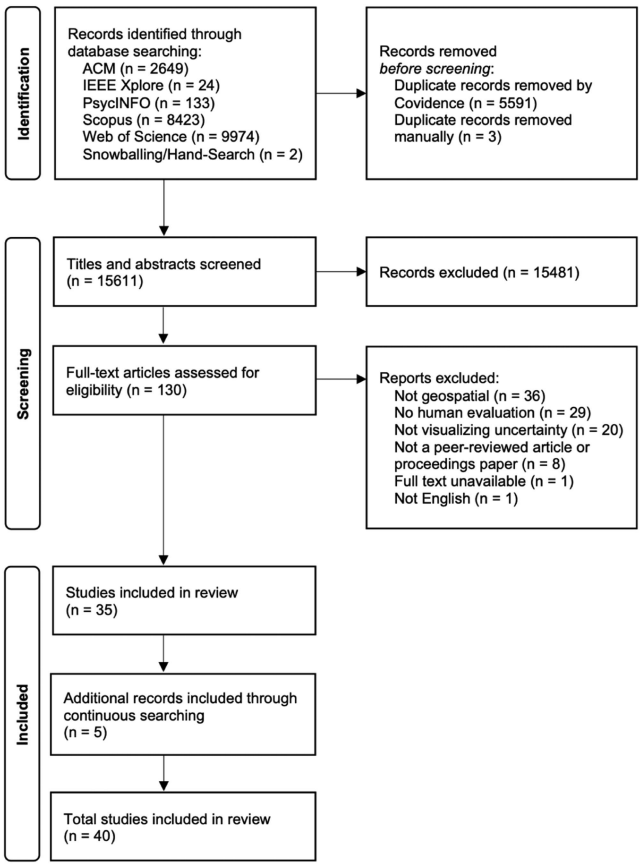


Fig. 1. Preferred reporting items for systematic reviews flow diagram depicting the literature search and eligibility screening outcomes.

from the full-text review either did not visualize geographical characteristics in a geospatial context, did not include HPP evaluation, did not visualize uncertainty information, were not a peer-reviewed article or complete proceedings paper, the full-text was unavailable, or the article was not available in English. In January 2024, the search was updated, and five recently published articles were included, totalling 40 included articles. The data extracted from the articles are included in the supplementary material, including a summary of the uncertainty visualization methods and the contexts in which they were applied (Table S2), and the general effect directions of the visualizations on various HPP metrics (Table S3), with synthesized results in Tables I, II, and supplementary tables (Table S4–S10).

A. *Overview of Study Characteristics*

The characteristics of the included studies are summarized in Table I. The number of published articles examining visualization techniques for geospatial uncertainty and their impact on HPP was 1-5 every three years until 2015. Between 2015 and 2018, the highest number of articles were published (35%, n = 14). The number of articles on this topic decreased to 6-7 every three years until the end of 2023. Author affiliations were from various institutions, with most articles from the United States and Germany.

TABLE I
SUMMARY OF ARTICLE CHARACTERISTICS

Article Characteristic		Studies (n=40), n (%)*
Study location	United States	20 (%)
	Germany	7 (%)
	Switzerland	3 (%)
	United Kingdom	3 (%)
	Australia	2 (%)
	Czech Republic	2 (%)
	Sweden	2 (%)
	Argentina	1 (%)
	Austria	1 (%)
	China	1 (%)
	France	1 (%)
	Italy	1 (%)
	Netherlands	1 (%)
	New Zealand	1 (%)
	Norway	1 (%)
	Spain	1 (%)
	Sri Lanka	1 (%)
Evaluation setting	Laboratory	23 (57.5%)
	Online	15 (37.5%)
	Field	1 (2.5%)
	Unspecified	1 (2.5%)
Study design	Quantitative	35 (87.5%)
	Qualitative	13 (32.5%)
	Mixed-methods	8 (20.0%)
Interface type	Static	28 (70.0%)
	Interactive	15 (37.5%)
Interface dimensions	2-dimensional	40 (100.0%)
	2.5/3-dimensional	1 (2.5%)
Map type	Multivariate	25 (62.5%)
	Univariate	24 (60.0%)
	Adjacent Maps	9 (22.5%)
Visualization technique	Colours	18 (45.0%)
	Shapes	14 (35.0%)
	Textures	6 (15.0%)
	Boundary regions	4 (10.0%)
	Symbols	3 (7.5%)
	Text	3 (7.5%)
	Grid noise	2 (5.0%)
Geospatial context	Weather	16 (40.0%)
	Position	14 (35.0%)
	Climate	5 (12.5%)
	Land cover	5 (12.5%)
Participant type	Novices	35 (87.5%)
	Experts	12 (30.0%)
	Experts & novices	8 (20.0%)
Evaluation measures	Performance	32 (80.0%)
	External decision	24 (60.0%)
	Perception	17 (42.5%)
	Performance & Perception	9 (22.5%)

*Percentages may sum to more than 100 due to overlapping categories.

Several types of geospatial uncertainty data were visualized. In the context of weather, the largest domain of uncertainty visualization techniques, applications ranged from visualizing hurricanes (15%, n = 6), precipitation in a specific region over time (8%, n = 3), and tornado risk (5%, n = 2) with a single article (3%, n = 1) on each of the following: wind speed, temperature, wildfire hazard, earthquake aftershock, and water-vapour mixing ratios. Positioning or location uncertainty was the second most extensive collection of articles in this review, where the visualization supported a user in navigating spatially by providing uncertain information about their location (15%,

TABLE II
UNCERTAINTY VISUALIZATION TRENDS ON HUMAN PERFORMANCE AND PERCEPTION

		Performance factor (M, SD)				Perception factor (M, SD)					
Technique	Map type	Decision-making	n	Response time	n	Understanding	n	Preference	n	Confidence	n
Boundary	Univariate	↓ (-0.72, 0.21)	3	↓ (-0.25, 0.25)	2	↓ (-1.00, 0.00)	1	↓ (-1.00, 0.00)	1	-	0
	Multivariate	↑ (1.00, 0.00)	1	↓ (-1.00, 0.00)	1	-	0	-	0	-	0
	All	↓ (-0.29, 0.77)	4	↓ (-0.50, 0.41)	3	↓ (-1.00, 0.00)	1	↓ (-1.00, 0.00)	1	-	0
Colours	Univariate	↓ (-0.28, 0.66)	12	↔ (0.00, 0.50)	8	↑ (0.25, 0.90)	16	↑ (0.13, 0.70)	10	↓ (-0.13, 0.78)	8
	Multivariate	↑ (0.08, 0.76)	12	↓ (-0.80, 0.40)	5	↓ (-0.07, 0.85)	15	↑ (0.25, 0.60)	12	↑ (0.75, 0.43)	4
	All	↓ (-0.10, 0.74)	24	↓ (-0.31, 0.61)	13	↑ (0.10, 0.89)	31	↑ (0.20, 0.65)	22	↑ (0.17, 0.80)	12
Grid noise	Univariate	-	0	-	0	-	0	-	0	-	0
	Multivariate	-	0	↔ (0.00, 0.00)	1	↑ (1.00, 0.00)	2	-	0	-	0
Shapes	Univariate	↑ (0.44, 0.68)	9	↑ (0.09, 0.51)	11	↑ (0.33, 0.60)	15	↑ (0.14, 0.83)	14	↔ (0.00, 0.00)	2
	Multivariate	↓ (-0.08, 0.86)	12	↔ (0.00, 0.00)	2	↓ (-0.11, 0.99)	9	↑ (1.00, 0.00)	1	-	0
	All	↑ (0.14, 0.83)	21	↑ (0.08, 0.47)	13	↑ (0.17, 0.80)	24	↑ (0.20, 0.83)	15	↔ (0.00, 0.00)	2
Symbols	Univariate	-	0	-	0	-	0	-	0	-	0
	Multivariate	↓ (-0.50, 0.50)	4	↓ (-1.00, 0.00)	1	↑ (1.00, 0.00)	1	↑ (0.67, 0.47)	3	-	0
Text	Univariate	↑ (0.25, 0.83)	4	↓ (-0.25, 0.83)	4	↑ (0.33, 0.94)	3	↑ (0.33, 0.00)	1	↑ (0.67, 0.47)	3
	Multivariate	↔ (0.00, 0.00)	3	-	0	↑ (0.67, 0.47)	3	↔ (0.00, 0.82)	3	↔ (0.00, 0.82)	3
	All	↑ (0.14, 0.64)	7	↓ (-0.25, 0.83)	4	↑ (0.50, 0.76)	6	↑ (0.08, 0.72)	4	↑ (0.33, 0.75)	6
Textures	Univariate	↑ (0.33, 0.00)	1	↔ (0.00, 0.00)	1	-	0	↓ (-1.00, 0.00)	1	-	0
	Multivariate	↑ (0.33, 0.47)	3	↔ (0.00, 1.00)	2	↑ (1.00, 0.00)	5	↓ (-0.33, 0.94)	3	↓ (-1.00, 0.00)	1
	All	↑ (0.33, 0.41)	4	↔ (0.00, 0.82)	3	↑ (1.00, 0.00)	5	↓ (-0.50, 0.87)	4	↓ (-1.00, 0.00)	1
All	Univariate	↓ (-0.01, 0.77)	29	↓ (-0.02, 0.56)	26	↑ (0.26, 0.81)	35	↑ (0.06, 0.80)	27	↓ (-0.08, 0.62)	13
	Multivariate	↔ (0.00, 0.76)	35	↓ (-0.50, 0.65)	12	↑ (0.23, 0.90)	35	↑ (0.23, 0.73)	22	↑ (0.13, 0.78)	8
	All	↔ (0.00, 0.76)	64	↓ (-0.17, 0.63)	38	↑ (0.21, 0.86)	70	↑ (0.14, 0.77)	49	↔ (0.00, 0.69)	21

(-) no data to report, *n* number of experiments conducted across all included articles, *M* mean trend value, *SD* standard deviation of mean trend value. The arrows represent the mean trend value for the visualization technique on the HPP factor. A downward arrow for response time should be interpreted as reducing response times. A horizontal line separates the individual techniques with the synthesized outcomes across all techniques.

n = 6). Otherwise, the context involved uncertain information about the land area when choosing an optimal location for placing an object or identifying where an object might be expected (20%, n = 8). Climate-focused articles presented visualizations about long-term changes in global temperatures (3%, n = 1), wind patterns (3%, n = 1), and precipitation (8%, n = 3), and articles that focused on aspects related to changes in the land or landscape included visualizations for the following: urban growth, soil depth, land porosity, vegetation, and general land change.

All of the studies in this review included two-dimensional visualizations. Static displays with non-moving images were more common than animated visualizations or displays that could be interacted with. Only one study included visualizations showing a third dimension but did not evaluate 3-dimensional (3D) data. Most experiments were conducted in a laboratory setting, while the others were conducted online. One study in the context of positioning uncertainty was considered a field study, where the participants used visualizations while navigating through a neighbourhood. Most studies used quantitative analyses, while some included qualitative analyses of interviews or post-study feedback. Fewer articles explicitly performed a mixed-method study, including quantitative and qualitative measures.

B. Uncertainty Visualization Methods

A summary of the included studies and the methods used for visualizing geospatial uncertainty are presented in Table S2 of the supplementary material. In addition to grouping by data type, the visualizations in each study were also grouped into univariate or multivariate categories. Univariate visualizations represented the uncertainty associated with a single variable in a geographic area. For example, univariate visual encoding included varying colour saturation, transparency, texture, or an outline of an uncertain area. Sometimes univariate visualizations were shown side by side: one with the variable being communicated to the user and an identical map juxtaposed displaying the uncertainty associated with that variable. Univariate visualizations depicted uncertainty by displaying multiple possibilities for a single variable, such as static ensemble lines or geolocation centroids.

Multivariate visualizations represented the relationship between two or more variables in a geographic area, where at least one of the variables was the confidence of another or their relationship. For example, multivariate visual encoding included combinations of techniques to represent these variables, such as colours and colour values, the shape and size of icons, and glyph sizes with colour values. In most cases, the multivariate visualizations appeared to be bivariate, representing the relationship between two variables in a geographic area, where one of the variables was the uncertainty of the other variable.

Whether the study examined univariate and/or multivariate visualizations, the techniques used for representing uncertainty varied. Colour was the most common technique to show uncertainty. However, it is essential to note that colours were used in different ways, including the degree of increasing lightness, whiteness, desaturation, or transparency to encode increasing uncertainty, using a spectrum of colours and mapping each colour to a discrete or continuous uncertainty value, or using two colours to represent uncertainty above and below a

threshold value. Shapes were the second most common technique, including rounded or polygonal. Shapes encompassed an area or region of uncertainty and often encoded uncertainty in a binary fashion, where the region within the shape indicated a particular uncertainty value. Some research also used shapes to encode uncertainty as a Gaussian distribution from a centroid. Similarly, the boundary technique also encompassed an area or region of uncertainty. These visualizations were distinctly different as they did not focus on a particular or necessarily recognizable shape or included centroids or Gaussian distributions within the shape area. A boundary region technique was often investigated in the context of how an enclosed region, which may not be uniform or include more than one boundary level, can be a less complex encoding method.

Textures overlaid a pattern or stippling onto a display. They may or may not purposely obscure the displayed values being communicated (e.g., to obscure high uncertainty areas). The uncertainty was encoded in a binary manner (e.g., texture vs no texture) or continuously as the texture becomes dense or sparse. Similarly, symbols were used in a couple of studies as an overlayed component in the visualization, indicating uncertainty that could be binary or continuous and could incorporate other techniques described using colour and texture. A couple of studies used text to describe uncertainty (e.g., "confidence low") or provide confidence percentages on a map grid. Finally, grid noise was the least explored technique in this review. The grid noise method involved increasing the width or graininess of the grid lines to indicate areas of increasing uncertainty continuously or discretely.

Colours, grid noise, symbols, textures, and text-based techniques were generally used to represent uncertainty in characteristics tied to defined geographic areas. Boundary regions and shape-based techniques were more commonly used to represent uncertainty that pertained to the exact geolocation itself, potentially including other characteristics at that location (e.g., the impacts of a hurricane).

C. Impacts of Multivariate & Univariate Displays

The effects of specific uncertainty visualization techniques for each article on HPP are presented in Table S3 of the supplementary material, with synthesized characteristics in Table II and Table S4–S10 of the supplementary material. While some articles included univariate and multivariate visualizations, only three studies explicitly set out to examine the differences between these map types. These articles used colour to encode the uncertainty [33], [34], [35]. In the multivariate map, one used colour whiteness [34], one used a colour saturation change [33], and one used a colour transparency change [35]. The univariate maps included the colour visualization technique in a separate map.

Among the three articles, Correll et al. [33] and Kubíček et al. [34] found that the multivariate map supported participants' understanding of uncertainty more than the univariate map. Confidence was high in the multivariate maps [34], [35], response times were quick [34], and decision-making was supported [33]. However, decision-making behaviours may have been more influenced by the display and not necessarily the multivariate colour [33]. In the context of informing the placement of ships that minimize danger, Correll et al. [33] identified that their specific modification to the multivariate map that suppresses high uncertainty values increased the likelihood of participants avoiding risky decisions.

Conversely, Schneider et al. [35] did not find the same benefits to multivariate maps in an earthquake aftershock prediction context for understanding uncertainty and decision-making outcomes. The authors found that adjacent univariate maps and a multivariate map using transparency did not support participants in recognizing that areas with higher uncertainty could experience worse aftershock outcomes [35]. Instead, adjacent displays showing the lowest and highest likelihoods of aftershocks supported participants in understanding that areas with greater uncertainty could see significant damage [35]. Interestingly, the authors did not label this visualization as showing uncertainty, and these adjacent visualizations showing the two extreme bounds of aftershock rates led to similar inaccuracies in reading the uncertainty levels compared to the multivariate map [35].

Other articles that included univariate or multivariate visualizations used shapes to represent uncertainty, focusing primarily on geolocation information [36], [37], [38]. These studies applied visualization techniques in the context of hurricane forecasts. They found that multivariate maps using animations of pixel trajectories supported the understanding of uncertainty and decision-making outcomes better than a static univariate display [38]. In the univariate context, static ensemble trajectory lines may better support decision-making and understanding of uncertainty than a multivariate uncertainty cone [36], [37], as the cone shape is more often mistaken as the hurricane growing [37]. Participants have also had greater misconceptions about the potential damage that could occur using the cone shape compared to ensemble lines [36].

Furthermore, while each of the visualizations across all techniques was categorized as univariate or multivariate when comparing these groupings against each HPP factor, there does not appear to be any strong trends (Table II). Multivariate may have faster response times overall, are possibly slightly more preferred, and may result in somewhat greater user confidence, but with high variability across the included studies. Trends for decision-making performance and understanding of uncertainty are similar between the two map types, with high variability across the studies. When considering all studies regardless of the categories, there were no trend directions for decision support or confidence, with moderate to high variability. The evaluated uncertainty visualizations also trend towards faster response times, supporting understanding of uncertainty and being preferred, with moderate to high variability. However, some trends emerge when looking at individual techniques (Table II) and when considering the contexts under which they were studied (Table S2–S10), with the caveat of synthesizing fewer studies.

D. Impacts of Visualization Techniques

1) Colours: The use of colour (Fig. 2) as an uncertainty visualization method was applied to several contexts and in many studies in this review [33], [34], [35], [39], [40], [41], [42], [43],

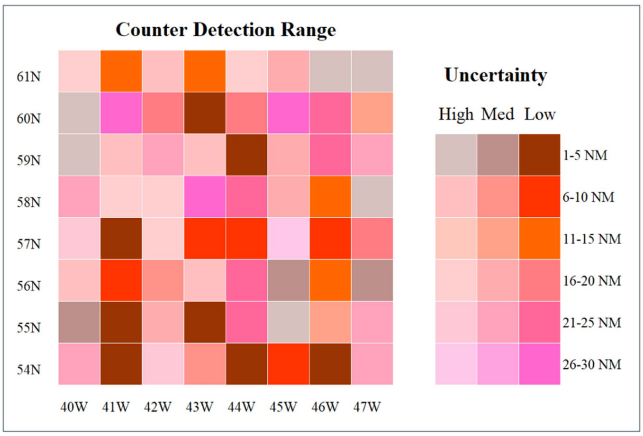


Fig. 2. Example of using value-based colours with varied hues in a multivariate display to communicate uncertainty.

[44], [45], [46], [47], [48], [49], [50], [51], [52], [53], including land cover for urban growth [45], soil depth [34], and land porosity for well drilling [40]; weather applications including wind speed [39], precipitation [43], [47], [50], temperature [51], tornado risks [46], [48], and earthquake aftershocks [35]; climate applications including global temperature changes [41], [44] and forecasted wind speeds over time [52]; and positioning with applications including the risk of an attack based on location [33], and object targeting [49].

Among the studies that explored univariate colour representations of uncertainty (Fig. 3), the approach appears to support users' understanding of uncertainty when designs use discrete colours [35], [41], [43], [44], [45]. However, one study that used only three discrete colour categories in a tornado warning context had a lower understanding of uncertainty and reduced trust in the visualization [46]. Univariate representations with continuous colour changes appear to have conflicting evidence, with some showing positive, negative, or no impacts on uncertainty understanding [33], [34], [39], [40], [45]. Using a spectrum-based (e.g., red, blue, green, etc.) or value-based (e.g., dark red, medium red, light red) colour change generally appears to support understanding of uncertainty in univariate representations [35], [39], [43], [44], [45]. However, the trends among the studies suggest that users may slightly prefer and have more confidence in value-based over spectrum-based univariate representations [35], [42], [45].

There is conflicting evidence for using univariate colour techniques in supporting decision-making. Regardless of being discrete or continuous, trends for univariate displays using value-based or spectrum-based techniques are generally downward with a moderate to high degree of variability among studies. For example, when identifying drilling locations, a univariate map using a continuous colour spectrum as the uncertainty representation may not support users in comparing or ranking locations well [40]. For monitoring wind speed and issuing a small aircraft safety advisory, univariate colour spectrum uncertainty maps may or may not reduce accurate decision-making when presenting error ranges at the expense of reduced reported confidence [39], with experts being less impacted. For three-colour categories representing different risk thresholds in a decision to take shelter from a tornado event, decisions were less accurate than a textual percentage of risk, potentially due to reduced understanding and trust [46]. In contrast, a spectrum-based univariate visualization may support decisions to leave amidst wildfire risk [42], and identify areas where aftershocks from an earthquake may occur [35].

For multivariate uncertainty maps using colours (Fig. 2), this approach had a small positive trend in supporting decision-making, yet with a slight negative trend in understanding uncertainty, with high variability among studies. There are also trends in reducing response times and positive trends in supporting user preferences [47], [48], [51], [52], and confidence [34], [50], [54]. Most of the multivariate colour visualizations used discrete colour categories and value-based colour changes to represent uncertainty, where there was a positive trend across the studies in user preference, confidence, and supporting decision-making while reducing response times. However, there was a negative trend in multivariate visualizations using colour value-based techniques to help the understanding of uncertainty with high variability.

Overall, colour spectrum-based visualizations were primarily used in univariate displays positioned adjacently, whereas colour value-based visualizations were mostly multivariate with discrete categories (Table S6). When comparing these colour visualization types, colour spectrum-based visualizations generally trended positively in supporting an understanding of uncertainty more than value-based visualizations. However, spectrum-based visualizations generally trended more negatively in user confidence and decision-making than colour value-based techniques. Colour value-based trended positively for user confidence with moderate variability between studies. There was also a small positive trend in decision-making performance for colour value-based visualizations with higher variability between studies. Both colour value- and spectrum-based uncertainty visualizations had reduced response times and positive trends for user preferences, with moderate variability between studies.

For visualizations that were categorized into the context of weather applications, there was a negative trend for colour in supporting decision-making and understanding of uncertainty with moderate and high variability, respectively (Table S7). In the context of position applications, there was a positive trend with high variability in supporting decision-making and understanding uncertainty among the few studies using this technique (Table S8). For the contexts of climate and land cover applications (Table S9 and S10), there were positive trends in using colour to support the understanding of uncertainty with lower variability among the few studies and positive trends for user preferences.

2) Shapes: The use of shapes (Fig. 4) was applied to location positioning and weather forecasting of a geographic location [36], [37], [38], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65]. Specifically, location positioning accounted for the most articles using shape as a technique, where applications were focused on supporting navigation uncertainty on a map through a mobile application [55], [56], [57], [66] or supporting target tracking in a sonar [62] and air traffic targeting

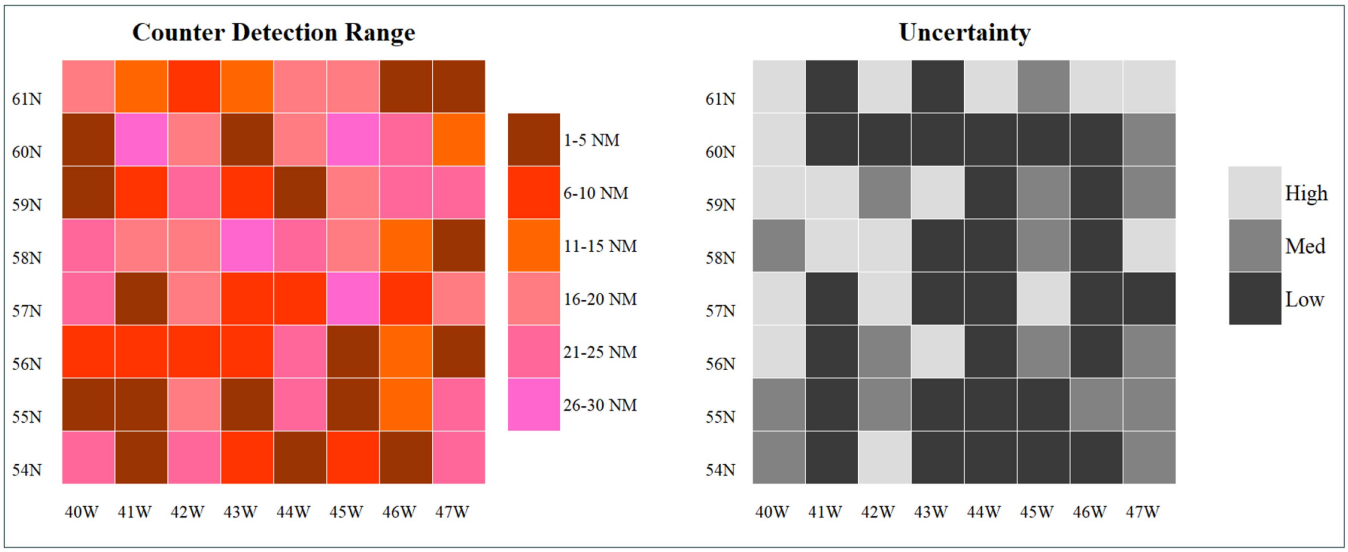


Fig. 3. Example of using spectrum-based colours (left) and value-based colours (right) in a univariate display to communicate uncertainty.

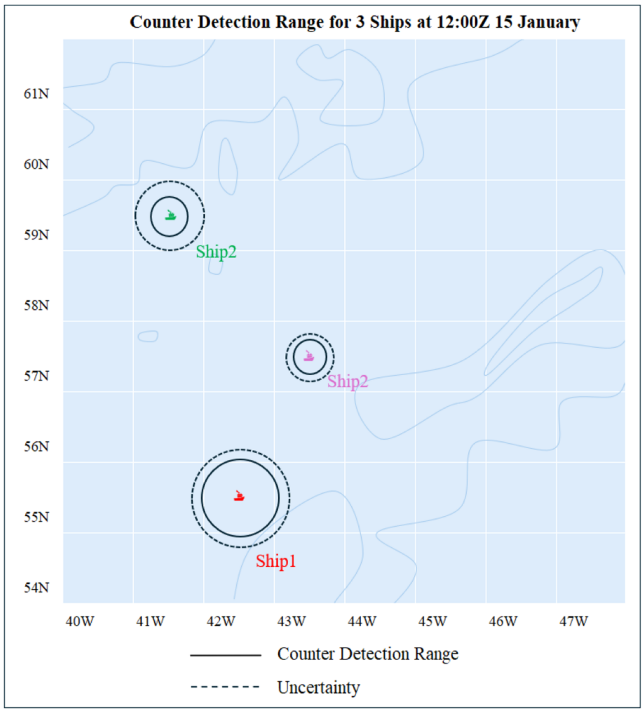


Fig. 4. Example of using shapes to communicate uncertainty.

context [60], [61]. Otherwise, representing uncertainty through shapes was used in hurricane risk applications [36], [37], [38], [63], [64].

In the context of using shapes to navigate, the most common method was using a circle of uncertainty with or without a centroid. However, some studies proposed new shapes for representing positional uncertainty on a map that included a "cloud" or highlighted the specific streets a user could be located on [55], [56], [66].

There was a positive trend toward an uncertainty circle with a centroid to indicate positional uncertainty being preferred [56], [58], [59], which also reduced the number of navigational mistakes a user could make in one study [59]. Providing context about the location by including landmarks also supported navigational accuracy while being a preferred way to provide supplementary information to help users orient themselves on the map [59]. When examining the impact of highlighting individual streets, the results on the preference for this method are conflicting. Some users preferred this method and reported reduced workloads [55], while others reported that this was their least preferable method compared to a circle or "cloud" representation [56]. The "cloud" shape was particularly useful in supporting users' understanding of higher uncertainty contexts [56]. However, there may not be major differences between these visual cues, like a transparent or faded circle with or without a centroid, in generally understanding positional uncertainty [56], [57], [59].

In a similar task situation, identifying targets on a map using shapes to visualize location uncertainty may be supported with visualizations that combine a circle with a centroid [60], [61]. In a sonar context, visually providing spatial uncertainty information may support decision-making accuracy for a decision-making task while reducing users' reported mental efforts, especially for novices [62]. However, there may not be a reduced amount of time required to complete the task regardless of expertise [62].

Shapes used to represent hurricane risk included cone, cone-centerline, ensemble trajectory (e.g., spaghetti plots) representations, and a similar technique showing pixel trajectories. All included studies involved a decision-making task regarding the potential damage a hurricane may cause over an area; however, there was no consensus on the impact on decision-making behaviours. One study found no change in protective decision-making behaviours when users were presented

with an uncertainty cone with or without a centreline representation of the hurricane's path and potential growth [63]. When users were tasked to estimate the level of damage from a hurricane when only seeing a centreline or a shape with a fuzzy or solid boundary (without a centreline), they had decreased damage judgments and reduced understanding of how the hurricane could grow [36]. Other research saw worse decision outcomes when the visualization included a centreline [37].

For supporting users in understanding increased potential damage from a hurricane, the evidence aligned more toward ensemble visualizations, which generally resulted in users estimating a greater risk of damage in different geographical areas [36], [37]. When the decision task concerns evacuations in response to an impending hurricane, pixel trajectories were more likely to support appropriate evacuation decisions than cone-centerline visualizations [38], [64]. These pixel trajectories (i.e., "zoomies") more broadly supported an improved understanding of potentially impacted areas and the storm's severity for novices when the pixel movement was animated and combined with a higher severity colour (e.g., red), creating a multivariate representation [38], [64].

When comparing univariate and multivariate shape-based visualizations, there was a stronger positive trend towards univariate representations supporting decision-making and understanding of uncertainty, with moderate to high variability across the studies. Multivariate techniques had a slight negative trend, with high variability among the studies regarding decision-making outcomes and understanding. However, when examining the types of multivariate shape visualizations, the downward trend is heavily influenced by the hurricane cone-type visualizations. In contrast, the animated pixel trajectories contributed to a positive trend in supporting decision-making and understanding uncertainty.

Overall, for visualizations that were categorized into the context of weather applications, there was a small negative trend for shapes supporting decision-making with high variability and no trend for understanding uncertainty with high variability (Table S7). In the context of position applications (Table S8), there was a positive trend with moderate variability for shapes supporting decision-making and understanding uncertainty. Shapes were not used in contexts involving climate or land cover applications.

3) Boundary Regions: The four studies that employed the boundary region visualization technique (Fig. 5) were applied to wildfire [42], sonar targeting [67], tornado risk [48], and wildlife contexts [49]. All studies included a decision-making task where the participants were required to respond using the uncertainty visualization beyond understanding the uncertainty representation.

For response times, there was a trend towards reduced response times using this visualization type to represent uncertainty in specific areas of interest regarding a potential weather event or object location/target identification task [42]. The small sample of studies provided conflicting results for decision-making performance, which had a negative trend with higher variability across the studies. Additionally, a boundary region

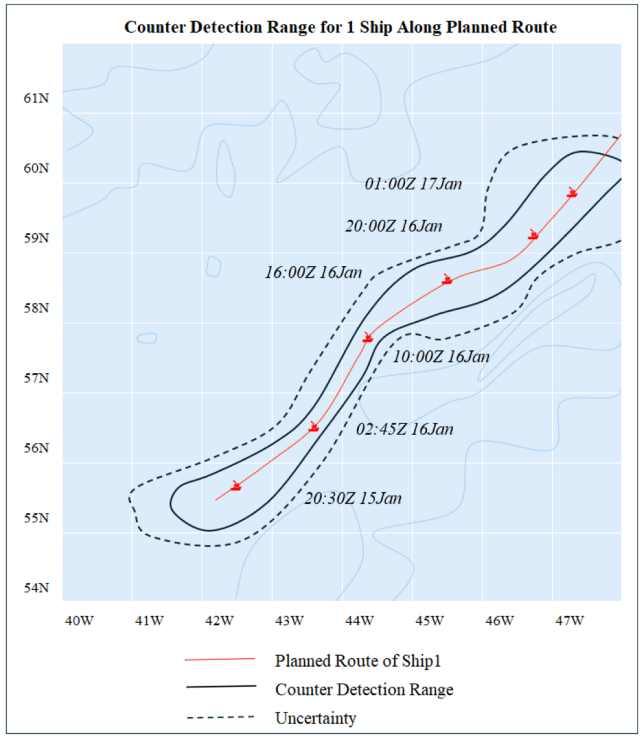


Fig. 5. Example of using boundary regions to display uncertainty.

may perturb users' attention toward specific areas of a display, which may or may not be beneficial for decision-making [67].

The different contexts also appeared to influence HPP when examining the impact of a boundary region. When choosing to shelter an aircraft amidst a potential tornado, a boundary region may not support an understanding of uncertainty or decision-making outcomes [48]. In wildfire evacuation, a boundary region visualization may be insufficient to support an accurate decision response [42]. However, it may increase user accuracy in object location tasks in non-weather related contexts when discrete levels of boundary region uncertainty are provided in a multivariate visualization [49].

Overall, for visualizations that were categorized into the context of weather applications, there was a strong negative trend for boundary regions supporting decision-making with low variability among two studies (Table S7). In the context of position applications (Table S8), there was a positive trend with high variability for boundary regions supporting decision-making among two studies, with a strong trend in reducing response times with low variability. Boundary regions were not used in climate or land cover applications.

4) Textures: Texture (Fig. 6) was applied in several contexts, including land cover for oil porosity [40], weather uncertainty regarding wildfires and temperature changes [42], global climate change with respect to precipitation [51], and location positioning in the context of landing a helicopter on sloped land [68]. All studies used this technique in the context of a multivariate map configuration except for one.

Different texture techniques were evaluated: transparency of texture, squares, lines, and dots. In general, across three studies,

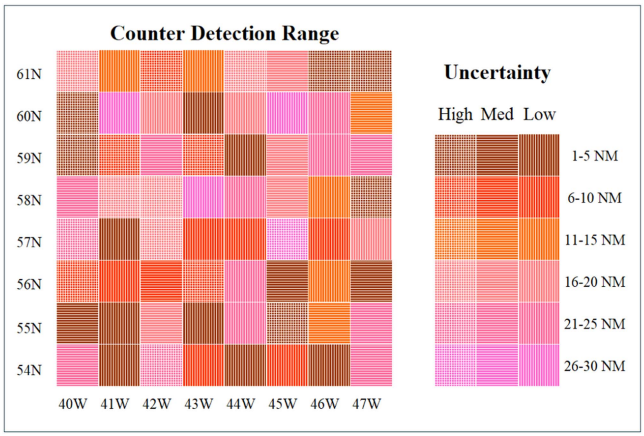


Fig. 6. Example of using textures to display uncertainty.

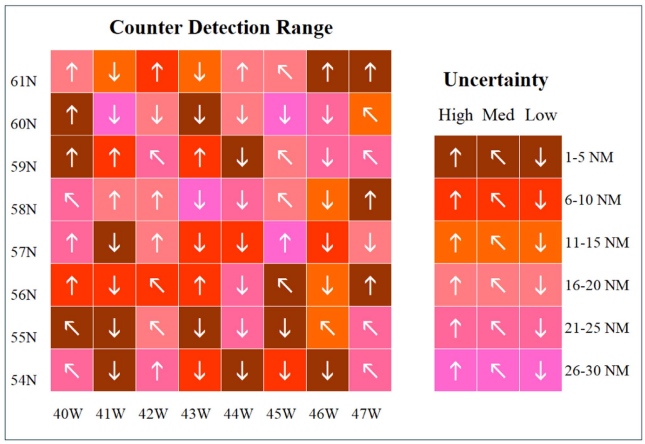


Fig. 7. Example of using symbols to display uncertainty.

texture can support an understanding of uncertainty without obscuring users' perception of the visualization in a multivariate map [40], [51], [69]. Johannsen et al. [69] specifically looked at intuitive perception and confirmed that participants perceived the increased density of a dot texture to be more certain than sparer dots, which aligns with some of the colour technique studies with lighter or greater transparency in colouring being intuitively more uncertain [34], [45], [51]. Retchless et al. [51] also identified that a dot texture was significantly preferred to squares or line textures.

However, using dots or stippling to indicate uncertain geospatial data in decision-making may not be ideal in other spatial decision-making contexts. When used to identify a safe location to land a helicopter when the slope of the land is uncertain in some areas, but while trying to land close enough to rescue a stuck hiker, decision-making accuracy was not impacted compared to visualizations without uncertainty. This technique increased decision-making time and impacted decision-making strategies, where users chose the safest landing that was further from the theoretical person needing rescuing [68]. However, this change in decision strategy may not directly relate to the technique used to represent uncertainty. In contrast, Cheong et al. [42] measured improved decision outcomes using a dot-based textural uncertainty visualization, but texture was one of the least preferred methods. There was conflicting evidence about whether this was the most preferred method among users, as another study identified a dot-based textural pattern as the preferred method [51].

Overall, for visualizations that were categorized into the context of weather applications, there was a strong positive trend for textures supporting understanding of uncertainty with no variability among two studies with three experimental outcomes (Table S7). There was also a negative trend for textures, which were users' preferred visual encoding, with high variability. In the context of position applications (Table S8), there was a positive trend with moderate variability for textures supporting decision-making among two studies. There was no observed trend in response times, with high variability. Textures were not used in climate or land cover application contexts.

5) Symbols: The use of symbols to represent uncertainty (Fig. 7) was used in three studies. The applications were for location and position contexts, including choosing the best locations for placing an airport [70] or a photo trap for wild animals [71], and in the context of precipitation uncertainty [47]. All visualizations were considered multivariate maps. Using symbols supported users' understanding of uncertainty [47], [70], [71] and were especially preferred with a fuzzy border [71]. However, using symbols was less likely to support decision-making tasks beyond this understanding [47]. Instead, symbols decreased performance or reduced rationality in decision-making in the context of fuzzy symbols despite being highly preferred [70], [71]. Decision-making behaviours were also perturbed with symbols with decreased decision-making rationality [70].

Overall, for visualizations that were categorized into the context of weather applications, there was no observed trend for symbols supporting decision-making with no variability among one study with two experimental outcomes (Table S6). There was a positive trend for symbols being the users' preference, with moderate variability among one study with two experimental outcomes. In the context of position applications among two studies(Table S8), there was a strong negative trend with no variability for symbols supporting decision-making. Symbols were not used in climate or land cover application contexts.

6) Grid Noise: Only two studies in this review examined the impact of noisy grid lines on HPP (Fig. 8). These studies were conducted in the context of land cover, including vegetation change [72], and general land change [73]. Incorporating grid noise within a multivariate visualization supported an understanding of uncertainty depending on the number of levels displayed and if the uncertainty was presented more uniformly rather than variable [72].

7) Text: The use of written text (Fig. 9) to convey uncertainty was applied in contexts including tornado warning forecast probability [46], precipitation severity uncertainty [50], and weather uncertainty regarding wildfire and temperature changes [42]. This technique was used in univariate and multivariate map configurations. Cheong et al. [42] and Gulacsik et al. [46] used univariate uncertainty maps. The decision-making context

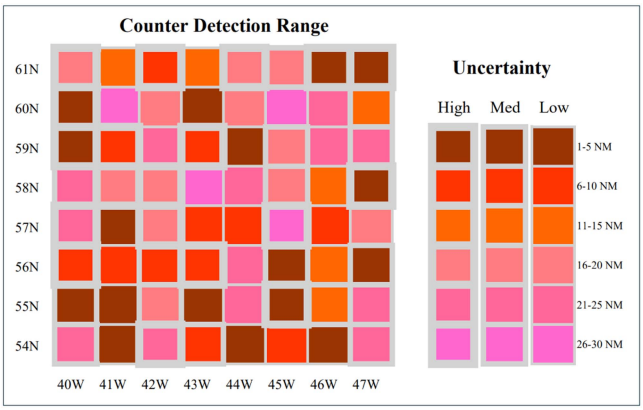


Fig. 8. Example of using grid noise to display uncertainty.

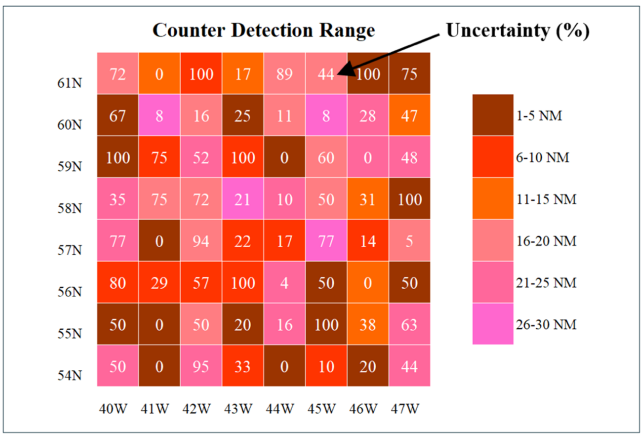


Fig. 9. Example of using written text to display uncertainty.

involved the participants' safety (i.e., evacuating their homes due to wildfire risk or taking shelter from a tornado). The authors found that representing uncertainty as a written percentage increased decision outcome accuracy, where participants took a longer time but made more protective decisions about their safety in response to a storm forecast compared to other visualization methods like colour encoding or written. Gulacisk et al. [46] also found that written percentages best supported participant understanding and trust of the uncertainty, especially in their more dynamic decision-making task where information was updated over time.

In contrast, the decision-making scenario used in the study by Clive et al. [50] involved selecting a day to take a walk outside when there was no severe weather forecast using a multivariate map configuration. This decision task involved lower stakes than the risk of potential death from the other weather events discussed. The authors' use of text written extrinsically to each area on the map to communicate the predictions found no difference in decision accuracy compared to colour value-based variations of border outlines and shapes yet was preferred and better supported participants' understanding [50]. However, written text on outlined regions resulted in lower confidence compared to the other techniques [50].

Overall, visualizations categorized into the context of weather applications showed a slightly positive trend for text supporting decision-making and preference with moderate variability (Table S6). There were stronger positive trends for text supporting understanding and confidence in weather applications with high variability and a strong trend in reducing response times with high variability. Text was not used in the other application contexts.

E. Performance & Perception Mismatch

Among all the studies in this review, there was no clear indication that preference for a visualization technique consistently improved decision-making performance. There were examples of studies in this review where, despite a higher preference for a visualization technique, performance decreased or remained unchanged [47], [48], [50], [71]. There were also examples where higher preference aligned with improved decision-making performance [42], [52], [74].

For understanding uncertainty, there was also no clear indication that preference for a visualization technique consistently resulted in an improved understanding of uncertainty information. There were examples of studies in this review where users had a higher preference for a visualization technique and a better understanding of uncertainty [44], [45], [51], [52], [65], [71]. There were also scenarios where users had a lower preference for a visualization technique but still had a good understanding of the uncertainty information [39], [50], [51].

IV. Discussion

A. Univariate & Multivariate Displays

The few studies that directly compared multivariate and univariate displays suggest that this visualization type may reduce response times, improve decision-making outcomes and confidence, and support understanding of uncertainty. Among the colour-based techniques, there appear to be some stronger trends in multivariate maps supporting confidence and reducing response times, especially when using value-based and discrete colours. The other studies included in this review do not explicitly examine the differences between multivariate and univariate uncertainty visualizations. However, when combining all the studies in this review, there do not appear to be any major differences in the trends between univariate and multivariate maps, suggesting that other aspects influence the HPP of uncertainty in geospatial visualizations, such as the variability in experimental designs and evaluation contexts. There may also be trade-offs between HPP of uncertainty for the designer to achieve the desired interaction experience, whether improved decision-making outcomes or improved abilities to accurately read the uncertainty values from the display.

For example, Schneider et al. [35] highlighted the importance of designing the visualization to optimize human performance outcomes in the visualization's application context, where the multivariate display underperformed. The visualization of the high and low likelihoods of earthquake aftershock areas in adjacent univariate maps traded off an explicit understanding

of uncertainty for an improved grasp of its implications, particularly the potential for much higher aftershock rates than forecasted [35]. This visualization focused on communicating the extremes of the forecast outcomes rather than simply the uncertainty itself, more effectively communicating potential surprises, where worst-case outcomes became more salient to users. Focused uncertainty communication, or direction of attention in particular ways, was also observed in visualizations such as unique shapes (e.g., clouds), boundaries, animations, and ensemble displays, making certain aspects of the information more salient. However, the saliency may or may not impact decision-making depending on the application.

Furthermore, the role of context is important when considering univariate or multivariate visualizations, as observed when using shapes to represent uncertainty in hurricane forecasts. While it appears that there are greater trends toward supporting decision-making and understanding of uncertainty for univariate shape visualizations in Table II, it is important to recognize that within these categorizations that were made, there are potentially useful techniques depending on the goal of the visualization. For example, supporting users in understanding where the centre of a hurricane may travel and making protective decisions appears to be better supported by visualizations that show the possibilities of trajectories, regardless of whether they are static ensemble lines or animated pixel trajectories [36], [37], [38], [64]. Similarly, a single trajectory line of the central location or a cone shape indicating the changing uncertainty over its width appears to lead to worse decision outcomes and lower understanding of uncertainty [36], [37], [38], [64]. Ultimately, it should be recognized that these categorizations of univariate and multivariate do not fully predict effectiveness for HPP; the specific design of the uncertainty visualization and its alignment with the users' task is crucial.

B. Visualization Techniques

1) Colours: There were differences in the specific techniques for the colour-based visualizations, including spectrum-based vs value-based and discrete vs continuous colours. It appears that colour spectrum-based visualizations can help users distinguish between different categories or levels of data and read the uncertainty information but may not provide a clear visual hierarchy [42], [75]. It may be more difficult for users to associate the meaning of different colours with different levels of uncertainty [75]. Conversely, the colour value-based visualizations may make it more challenging to read the uncertainty information between lightness, darkness, or transparency levels, which may partly occlude the data but provide a clearer hierarchy of meaning (e.g., darker is more certain, lighter is less certain). The colour-value-based uncertainty approach may be more easily separated from the underlying data visualization. Notably, this separation between the data and its uncertainty using different techniques is arguably more pronounced in the texture-based approach, which also showed higher trends in understanding and decision-making support (Table II).

Discrete colour visualizations may also lead to greater confidence and preference than continuous colour visualizations, possibly due to how people process visual information and how these relate to decision-making. While discrete and continuous colour visualizations may show negative trends in decision-making outcomes, discrete methods were slightly better. Discrete colour visualizations simplify the visual field by quantizing the data into a limited set of distinct colours with clear boundaries [76], potentially aligning better with how people naturally categorize information. These categorizations may lead to reduced errors in perception compared to determining the precise value along a continuum [33]. However, the trade-off between exactness and clarity needed for the visualization will depend on the specific context and the decisions a user is expected to make based on the information being displayed [33].

2) Shapes: For the shape-based visualizations, there was a trend towards univariate representations that better supported decision-making and understanding uncertainty than multivariate representations. While it is important to realize the context in which these visualizations were applied, ranging from estimating one's geolocation to the potential risk of hurricane damage in different regions, the univariate representations may be more effective because of their simplicity. For example, while the cone of uncertainty is intended to show the range of possible storm paths, it is often misinterpreted as representing the size or intensity of the storm, rather than the width of the cone representing how location uncertainty changes over time [64]. The increasing size of the cone may also be more salient to users, biasing their attention towards certain areas [11].

Conversely, univariate shapes may have less complexity. While univariate visualizations can similarly generate saliency through the use of bold outlines or including other visual cues like size and colour [11], [77], [78], these features may not be as meaningful as they are in shapes like the uncertainty cone in the context of a hurricane. Indeed, the subtle differences in some visual cues presented for positional uncertainty (e.g., circles with or without a centroid) did not substantially alter users' perceptions of uncertainty. However, animating the ensemble pixels for hurricane paths, where a time variable is directly conveyed through visual movement, is one multivariate shape technique that can support decision-making and understanding positional uncertainty, de-emphasizing the potential misconceptions of size or growth. Additionally, this type of multivariate shape is another example of the separation between the data (i.e., position over time) and its uncertainty (i.e., the ensemble of possible routes) being more pronounced compared to the uncertainty cone, as discussed in the context of the multivariate colour value-based technique.

3) Boundary Regions, Grid Noise, Symbols, Text, & Textures: For the other visualization techniques, fewer studies comprehensively evaluated HPP of uncertainty compared to studies using colours and shapes. While trend directions for these techniques are included in Table II, more research is needed to validate and verify some of the emerging trends within various contexts.

C. Performance & Perception Evaluation

In this review, there was significant heterogeneity in the measurement of HPP of uncertainty visualization techniques. The

studies used a range of tasks, from simple map-reading exercises to complex decision-making scenarios. Some tasks focused on identifying the level of uncertainty, while others concentrated on how uncertainty affects choices or resource allocation. Many studies assessed this factor for decision-making performance through the accuracy of choices in an external task, often in scenarios with a correct or optimal response given the uncertain information. These tasks may have had greater negative personal outcomes with poorer decision-making. For example, getting caught in a potential wildfire event [42]. Other decision-making tasks may not have had significantly negative personal consequences, such as deciding the best place to put a photo trap for monitoring wildlife [71]. This variability in tasks makes direct comparisons across studies challenging.

There was also variation in the experimental context in which the visualizations were tested, from general spatial decision-making to more specific scenarios like hazards, weather forecasts, and military operations. These studies were conducted in different testing environments, from the laboratory to the field and online. These experimental context differences resulted in different data collection methods, including qualitative interviews, questionnaire-based approaches, quantitative experimental designs, or a combination of approaches to measure response times and subjective measures like understanding, preferences, and confidence. Additionally, the participants' expertise varied, including novices, experts, and sometimes a mixture of these user populations, which was not always explicitly defined by the authors. While there are some consistent metrics, like response times, the varied tasks, contexts, and measurements introduced complexities for how uncertainty visualizations are understood and used in decision-making. There is a clear need for more standardized and unifying approaches to better understand HPP across diverse user types and to enable more direct comparisons in more geospatially generalizable ways. This may aid in developing clearer guidelines that can help inform the design of effective uncertainty visualizations.

D. Performance & Perception Mismatch

Simple preferences toward specific uncertainty visualization techniques did not always equate to improved performance outcomes [47], [71], [79]. The studies in this review also indicated no straightforward relationship between liking a visualization method and enhancing decision-making or understanding. This insight is important because it challenges the idea that user preference should be a primary factor in choosing a visualization method. Instead, it suggests that other factors, perhaps cognitive or contextual, might play a more critical role in the effectiveness of a visualization technique.

Additionally, this review suggests a key difference between comprehending uncertainty and effectively using it for decision-making [4], [70]. Visualizations might successfully convey that uncertainty exists and support users in understanding its level, but not necessarily how to integrate this into the decision-making process. Indeed, MacEachren et al. [4] note that uncertainty visualization research appears to assume that visually depicting uncertainty is useful for decision-making. For example, in Scholz et al. [47], participants showed that they understood which symbols represented higher uncertainty, but it did not necessarily translate into better choices for crop placement. Similarly, Brus et al. [71] focused on symbol intuitiveness but found that user understanding and preferences did not necessarily lead to choosing more suitable locations for placing a wildlife photo trap.

E. Expert & Novice Users

Most of the studies in this review focused on novices, individuals who are often characterized as non-professionals, distanced from science, or lacking certain knowledge or skills [80]. However, it is essential to note that within this category, the term novice was not used uniformly, including students and the general public. The term novice can be ambiguous in its use in visualization evaluation studies [80]. Subsequently, it has been noted that groupings within the novice category may not generalize, especially with small sample sizes [80]. In this review, students and the general public were considered novices relative to professionals or participants with substantial experience who were considered experts. It is important to recognize where nuance may exist and be careful with over-generalizations between these categories. For example, students who participated in the included studies may have a stronger grasp of probability theory than the general public [46], or assigned themselves to being a novice user or very experienced [70]. In addition, some members of the general public may be highly educated and report a strong familiarity with the type of geospatial graphics being evaluated [50].

In a previous review by [13], the authors identified the importance of distinguishing between two types of expertise when performing human experiments about uncertainty visualizations, as experts may be no better than novices when extracting information from a visualization; however, an experts' prior experiences may inherently support their decision-making performance. The authors proposed distinguishing between two levels of expertise related to understanding the visualization and making decisions under uncertainty. Unfortunately, we identified few studies directly evaluating the impacts of uncertainty visualization techniques on improving decision-making performance between novices and experts or making these distinctions clear. The few studies included in this review suggest that novices and experts can understand uncertainty visualizations [45], [47], [70], [71], [72], [73]. In Tables S4 and S5 of the supplementary material, when collapsing among all visualization types in this review, there does not appear to be any notable differences in trends for decision-making and response times or preferences and confidence between novices and experts. However, the understanding of uncertainty by experts had a stronger positive trend than novices, which may be inherent because of greater knowledge and experience [13]. For novices, there is limited evidence that texture-based techniques may better support decision-making and understanding of uncertainty but may not be preferred or support confidence (Table S4).

Future research can further inform the design of uncertainty visualizations for novices and experts in the context of geospatial information. First, experiments with uncertainty visualizations should explore how decision-making can be supported in different contexts, such as by factors that better support confidence [81], or that support saliency and understanding of the information needed for optimal decision-making [35]. Second, as this review identified, most visualizations are on static displays. Recent studies have seen positive impacts with animated and interactive displays with novice users [38], [46], [50], [64], and when they have control over the threshold of uncertain information to display for decision-making [52]. There is an opportunity to explore uncertainty visualization and the potential for adaptive interfaces to improve both understanding uncertainty and making decisions, depending on the task and the user's expertise, while being mindful of cognitive resource limitations.

F. Visualization Contexts

In this review, all articles that included human participant evaluations had two-dimensional (2D) data representations, most of which were static visualizations. The one article that included a 2.5/3D visualization did not evaluate the 3D data [40]. Future display technologies may incorporate mixed and virtual reality with 3D components in a dynamic, time-series context. Therefore, it is essential to determine how the current methods for visualizing uncertainty may apply in this context or if they need to be adapted to support the HPP of uncertainty information. Having an additional dimension also enables new ways to present uncertainty. For example, instead of using the z-axis for depth, its use in conveying uncertainty could be explored. 3D uncertainty visualization research exists [82], [83], yet no articles were found that met the inclusion criteria for this review. Further research should investigate the impact of 2D uncertainty visualization techniques and modified or new techniques on HPP measures in 3D displays.

Furthermore, only a few studies in this review explored interactive and animated displays [38], [46], [64]. This included an example where an interactive visualization supported HPP factors when users were given control to change the displayed uncertainty thresholds [52]. Similarly, animated displays also resulted in decision-making benefits [38], [64]. Indeed, interactive geospatial displays can help users filter information, reduce display complexity, and support users in decision-making [84]. Given the potential balance of risks and benefits to user control over displaying uncertainty, further exploring this factor, its feasibility, and its potential impact in different contexts and task scenarios is needed.

Finally, in-depth studies evaluating the context of use and workflow integration of potential uncertainty visualization designs were also limited in this review, which included several laboratory and survey evaluations. This limitation is a significant concern when considering how uncertainty visualizations may impact decision-making performance positively or negatively in the real world [73]. More research is needed that considers workflow integration of uncertainty visualizations, including the type of data represented, the experience of the user(s) performing the task, the decision-making process within which the uncertainty is incorporated, time limitations, and the potential consequences of decision-making.

G. Potential Research Directions

Prior reviews on visualizing geospatial uncertainty emphasize that visualizations should support a user's needs and tasks [28], [29], highlighting the critical importance of understanding the work domain. Our review builds on Kinkeldey et al. [13] by expanding on and exploring some of the authors' proposed evaluation categories for the included studies while recognizing that the objective of this review was limited in making comparisons between study findings. Despite including studies that explore similar techniques in similar contexts, our review helps identify promising techniques while recognizing the nuances of the varied contexts, users, and outcomes being evaluated.

Based on the heterogeneity of the findings, there is a need for more research in the context of uncertainty visualization for geospatial data related to the environments explored in this review. To support the integration of uncertainty visualizations into future display technologies, future research could explicitly explore the design and evaluation of uncertainty visualizations from a work domain perspective that can improve the coupling between the perception of uncertainty and HPP. One such methodology is a design process called Cognitive Work Analysis, which emphasizes understanding work domain constraints to embed skills- rules- and knowledge-based information within visualizations to support decision-making effectively [85], [86]. Insights from the models produced through this analysis inform an ecological interface—a design technique which has yet to be explored in the use of geospatial uncertainty for decision-making tasks—which has explicit advantages in affording users the ability to achieve a tighter perception-action response in decision-making [87], [88]. Developing a deep understanding of the domain and decision-making process can inform visualization techniques that are more likely to support improved human performance by helping users understand the relationships between the uncertain information and potential decision outcomes or the degree to which the uncertain information should be considered or highlighted in different tasks.

H. Limitations

The first author, who is an experienced researcher on two prior human factors-related scoping review articles, screened and extracted the data for this review under the guidance of the second author with greater domain expertise [89], [90]. The authors met to discuss the articles and preliminary findings, where the second author also critically examined the plausibility of the review outcomes to support the rigour of the analysis. The initial results were also presented at the annual research meeting among defence scientists working on display design technology for national defence strategies. Overall, this process supported the efficiency of the review; however, the single-screener methodology may have introduced bias and risk of error through missing relevant studies [89], [91]. Given the

total number of articles screened and the heterogeneity of the results, we are confident that the limitation of single-author screening would not significantly alter the general findings and recommendations for future research directions.

This review does not extend beyond geospatial information or applications beyond geographic uncertainty data. Other contexts where uncertainty information visualization is being explored may provide similar or differing impacts than the settings included in this review. For example, a recent systematic review has identified that information visualization can support decision quality and response times, but depicting uncertain information can reduce decision confidence [30]. Our review explores uncertainty visualization techniques while highlighting the importance of context on HPP outcomes, including the task, the visualization approach, and the user type. While insights from other domains could be transferable, this transferability was outside the scope of this review. We recommend that future research explores and critically examines potential cross-domain overlap on human performance and perception of uncertainty visualization techniques.

Other information included within certain visualization approaches that were explored in the included articles may also impact HPP of uncertainty, but these details were not extracted or explicitly considered in this review if they were not directly related to uncertainty as this was outside the scope. However, this additional information is another contextual factor that should be considered in future research studies to understand optimal designs and use cases for these multivariate displays. Additionally, the categorizations used for univariate and multivariate visualizations, application contexts, and visualization techniques were not perfect in creating groupings based on similarities. This was partly a limitation in the comprehensiveness and details provided by the included articles and the heterogeneity within the various applied approaches and methods. Further research can support future reviews through standardized reporting.

Finally, while this review synthesizes trends in the effects of uncertainty visualization techniques on HPP of uncertainty, these trends should be interpreted cautiously. The directionality of the effects was derived from individual evaluations with comparisons made between visualizations with and without encoding uncertainty, as well as between different uncertainty visualization techniques—some of which came from the same studies—and assigned numerical values for synthesizing. This process inherently involves a level of subjectivity in interpretation. Additionally, the variability in reported findings, as reflected in the means and standard deviations, highlights that the observed trends are not absolute but indicative of patterns in the reviewed literature. Given the nature of this review, designers and researchers should consider these findings as exploratory insights rather than prescriptive guidelines, ensuring that any application of these trends aligns with the specific context, task, and user population. Future work should further investigate these relationships through experimentation and real-world evaluations to refine the understanding of how uncertainty visualizations affect the HPP of uncertainty in geospatial contexts.

V. Conclusion

This comprehensive review highlights the complexity of geospatial uncertainty visualization, revealing a fragmented research landscape with diverse contexts and varying HPP outcomes. While some trends emerged among the included studies, considerable uncertainty remains regarding the most or least effective methods for representing uncertainty in geospatial data, where context appears to be an essential factor. Additionally, much of the existing research focuses on supporting novice-type users in decision-making tasks, with fewer studies examining how domain experts or professional users interact with these visualizations.

Despite the increasing use of familiar visualizations such as colours and shapes, this review highlights variations of HPP outcomes in different contexts. Integrating the intended human performance and/or perception outcome(s), the context within which the visualization is being designed, and the emerging trends of this review can help guide design choices to support user interpretation and action on uncertainty.

Future research should further explore contextual factors that influence the perception and use of different uncertainty visualizations in geospatial settings informed by emerging trends, assess techniques within real-world workflows, and examine the potential for established interface design frameworks to enhance practical applicability. Addressing these gaps will advance the field and provide domain professionals with more effective decision-support tools for navigating complex geospatial uncertainty data.

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